Rajagopal Somaskandan

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Professional Summary

Applied Machine Learning Engineer and IEEE-published researcher (M.Sc. CS, 2027) with experience building **production-grade AI systems**. Delivered high-impact models including fraud detection (94% recall, ROC-AUC 0.995) and real-time ML systems, with end-to-end ownership from training to FastAPI deployment.

Technical Skills

Languages: Python, Java, SQL
Machine Learning: Scikit-learn, XGBoost, LightGBM, PyTorch, SHAP
NLP & GenAI: BERT, Sentence-Transformers, spaCy, NLTK, LangChain, ChromaDB, FAISS, RAG
Computer Vision & 3D: OpenCV, Optical Flow, COLMAP, 3D Gaussian Splatting, FFmpeg
Databases: PostgreSQL, MongoDB, Redis, SQLite
Frameworks & Tools: FastAPI, Flask, Streamlit, Docker, Git, GitHub, Netlify, Render
Data Analysis: Pandas, NumPy, Matplotlib, Plotly, Power BI

Education

Sathyabama Institute of Science and Technology — **B.Sc. Computer Science (AI Specialization)** | First Class — *2025*
Sathyabama Institute of Science and Technology — **M.Sc. Computer Science** | — *Expected 2027*

Experience

Production AI Developer (Intern) | *InnoXR Labs Pvt. Ltd.* — *Mar 2026 – Jun 2026*
- Developed and deployed ML-powered FastAPI microservices for production AR/immersive applications, enabling scalable API-first AI integration.
- Owned end-to-end feature delivery — from model development to API deployment — contributing directly to production releases in a fast-paced startup environment.

Data Analyst Intern | *India Advocacy* — *Jul 2024 – Oct 2024*
- Built Java + Selenium pipelines to scrape and process ***1M+ records***, transforming raw data into ***15–20k high-quality datasets*** through multi-stage validation.

Security Trooper (National Service) | *Singapore Armed Forces* — *Jan 2020 – Jan 2022*
- Supported security operations, operated authorized transport vehicles under regulated operational environments and assisted Ministry of Health COVID-19 task force initiatives through structured data collection and reporting

Sales Associate (Part-Time) | *Bata* — *Feb 2022 – Aug 2022*
- Managed high-volume customer interactions, delivering efficient service in a fast-paced retail environment.
- Handled billing, inventory coordination, and issue resolution, improving in-store operational efficiency.

Projects

FinGuard Pro: Real-Time Financial Fraud Detection & Compliance System — *IEEE Research Project (ICAAIC 2025)* — *2025*
XGBoost, SHAP, FastAPI, Streamlit, SQL
- Built an end-to-end fraud detection platform with real-time scoring, REST APIs, and role-based dashboards for transaction monitoring.
- Achieved **94% recall** and **ROC-AUC 0.995** on imbalanced financial data; integrated SHAP for **explainable, audit-ready decisions**.
- Developed admin/user dashboards with automated **PDF/ZIP fraud reports**, enabling scalable investigation workflows.

CareerGenie: AI Career Intelligence Platform (RAG + LTR + Multi-Agent) — *Hackathon Winner (Origin 2026)* — *2026*
FastAPI, React, Ollama (LLaMA 3.2), ChromaDB, Sentence-Transformers
- Built an end-to-end career pipeline integrating resume parsing, ATS scoring, semantic job matching, mock interviews, and personalised learning roadmaps.
- Implemented RAG-based job matching (MiniLM + ChromaDB) with adaptive scoring and **BPR learning-to-rank**, improving relevance using user feedback signals (EMA updates).
- Designed a **multi-agent debate system** (propose–critique–synthesise) with confidence-gated validation to improve output reliability and reduce hallucinations.
- Integrated local-first LLM stack (Ollama) with cloud fallbacks (Groq/Claude/Gemini), enabling **offline-first inference** with scalable FastAPI backend.

MonoSplat: Single-Video 3D Reconstruction Pipeline — *Internship Project* *2026*

PyTorch, FastAPI, COLMAP, FFmpeg

- Built a pipeline converting **single RGB video into real-time 3D Gaussian Splat scenes** viewable in browser.
- Implemented **Structure-from-Motion (COLMAP)** for camera pose estimation and PyTorch-based Gaussian Splat training.
- Designed **hybrid CPU–GPU architecture**: local preprocessing (FFmpeg + SfM) with cloud GPU training (Colab/Runpod).
- Developed FastAPI backend with async job pipeline and **Three.js viewer achieving 30+ FPS** real-time rendering.

S.A.F.E: Real-Time Crowd Anomaly & Risk Detection — *Design Thinking Project* *2026*

OpenCV, Farneback Optical Flow, MAD, FastAPI, Streamlit

- Designed a privacy-preserving anomaly detection system using zone-based optical flow aggregation for real-time CCTV monitoring without storing raw footage.
- Reduced false positives by **40–60%** via global motion compensation, eliminating a key failure mode in drone and handheld deployments.
- Achieved **ROC-AUC 0.8391** on cross-domain Mall dataset with **zero retraining**, prioritizing generalization over dataset-specific tuning.

Redline: Multilingual Preventive Healthcare Chatbot — *Smart India Hackathon (SIH 2025)* *2026*

IndicBERT, Groq LLaMA 3.3, FastAPI, Telegram API

- Engineered a hybrid AI system combining IndicBERT (10+ languages) with LLM fallback, achieving **82.4% intent accuracy**.
- Built **confidence-aware routing** and rule-based medical safety layer, achieving **95.6% safety filtering accuracy**.
- Developed India-specific healthcare knowledge base with real-time outbreak alerts across 28 states via Telegram deployment.

SourceUp: Explainable Supplier Recommendation System — *Final Year Capstone* *2026*

Sentence-BERT, FAISS, FastAPI, Scikit-learn, Selenium, Java

- Built a supplier intelligence pipeline using **Java + Selenium** for large-scale web scraping and data collection.
- Implemented semantic retrieval using **Sentence-BERT + FAISS** for natural language supplier search.
- Designed a **constraint-aware scoring engine** incorporating price, delivery time, and reliability via penalty-based optimisation.
- Implemented an **explainability layer** with feature-level insights and risk classification for transparent supplier selection.

Publications & Conferences

FinGuard Pro: Explainable AI for Financial Fraud Detection | *ICAAIC 2025* *ISBN: 979-8-3315-6587-9*

Real-Time Explainable Fake News Detection using BERT-LSTM | *ICASET 2025*

Achievements & Hackathons

Industry Innovation Hackathon | *Sathyabama Institute of Science and Technology — National Level* *Mar 2026*

- Won ***Best Industry Innovation Award*** at Industry Innovation ’26 (organized with ATRIBS and Bluewind Innovations) — defeating 63+ teams from institutions across India with the CareerGenie AI platform.

Origin 24-Hour Hackathon | *SIMATS Engineering, Chennai* *Apr 2026*

- Competed in a 24-hour sprint organized by the SIMATS Hackathon Club, Saveetha Institute of Medical and Technical Sciences.